

# R1 Project Demo

## *Neighborhood Tool Sharing*

### **Team Members**

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# Overview of the Domain Model and ER diagram

## 8 Entities, 5 Core Relationships

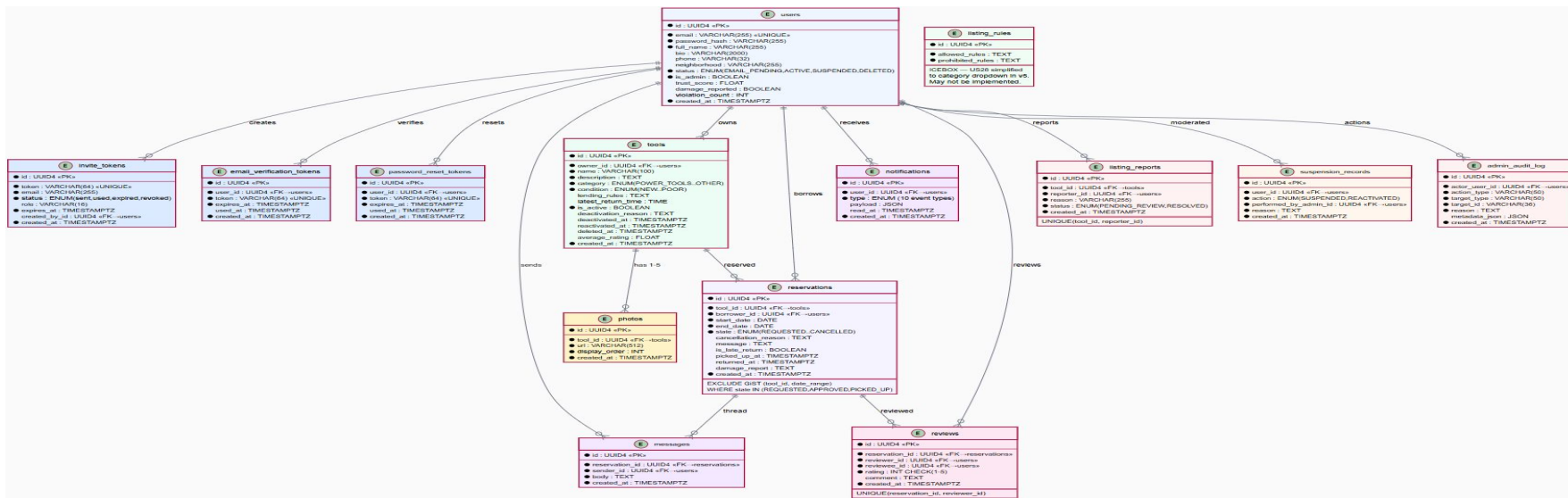
User, Tool, Reservation, Review, Invite, Photo, Notification, AdminAuditLog

- **User owns Tools** • User borrows via Reservations
- **Reservation → Review** (one per party)
- **Tool → Photos** (1–5 scale limit)

## Key DB Constraints

- **GIST EXCLUDE** on (tool\_id, daterange) — strictly prevents any double-booking
- **UNIQUE** (reservation\_id, reviewer\_id) — limits to one review per party
- **Soft-delete** with PII anonymization & active-reservation guard

**Enums:** 5 ToolCategory · 5 ToolCondition · 6 ReservationState · 14 NotificationType



# Major Accomplishments — Backend

## Backend

### 46 Endpoints / 7 Routers

Fully Implemented & Tested

Comprehensive coverage of core services ensuring robust data layers and business logic.

#### Endpoint Breakdown:

- **Auth:** 13 endpoints
- **Tools:** 11 endpoints
- **Reservations:** 10 endpoints
- **Reviews:** 5 endpoints
- **Admin:** 4 endpoints
- **Notifications & Health:** 3 endpoints

## Infrastructure Highlights

- **JWT Security:** Access & refresh tokens with automatic rotation.
- **Rate Limiting:** Configurable active protection on all auth endpoints.
- **Double-booking guard:** GiST EXCLUDE at DB level and app-level overlap check.
- **APScheduler:** Automated stale cancels, escalations, and cleanups.
- **Photo Uploads:** Magic-byte validated with a 5 MB disk limit.

### Neighborhood Tool Sharing API 0.1.0 OAS 3.1

[OpenAPI Spec](#)

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#### auth

|        |                                  |                     |
|--------|----------------------------------|---------------------|
| GET    | /api/v1/auth/invites             | List Invites        |
| POST   | /api/v1/auth/invites             | Create Invite       |
| POST   | /api/v1/auth/register            | Register            |
| POST   | /api/v1/auth/verify-email        | Verify Email        |
| POST   | /api/v1/auth/resend-verification | Resend Verification |
| POST   | /api/v1/auth/login               | Login               |
| POST   | /api/v1/auth/refresh             | Refresh             |
| POST   | /api/v1/auth/logout              | Logout              |
| GET    | /api/v1/auth/me                  | Get Me              |
| PUT    | /api/v1/auth/me                  | Update Me           |
| DELETE | /api/v1/auth/me                  | Delete Me           |
| POST   | /api/v1/auth/forgot-password     | Forgot Password     |
| POST   | /api/v1/auth/reset-password      | Reset Password      |

# Major Accomplishments — Frontend

## Core Member Workflow

- Built R1 member-facing pages for login, registration with invite token, email verification, forgot/reset password, profile setup, and edit profile.
- Added mock authentication behavior using frontend state/localStorage so the R1 demo can show login/logout and protected member navigation.

## Tool Listing Workflow

- Implemented major tool-sharing pages: Browse Tools, Tool Detail, Create Tool Listing, and Edit Tool Listing.
- Added frontend validation for required fields, photo upload limits, duplicate tool names, latest return time format, and HST date/time labels.
- Added owner-side delete/deactivate listing UI, including active-reservation blocking and confirmation behavior.

## Reservation + Review Workflow

- Built reservation workflow UI for requesting a tool, owner approve/deny/cancel actions, borrower cancel, confirm pickup, and confirm return.
- Added returned-reservation review support, including Leave Review flow and Review History page.
- Added frontend reservation conflict messages and future backend-ready conflict wording for overlapping reservation dates.

## Notifications + Admin Support

- Improved the Notification Center with unread/read filters, notification badges, Mark as Read, Mark All as Read, and Dashboard notification sync.
- Added basic admin/demo pages, including Admin Invite Management and Admin Listing Controls.
- Added R1 support for overdue pickup / auto-cancel notice UI to show how stale approved reservations will be handled.

# Major Changes to requirements

Some User Stories have changes

**New User Story:** Admin Invites a New Member

- Priority is **high**. Included in R1 milestone.

**Iceboxed User Stories:** User Story 28 (Tool Categories) & User Story 33 (Admin Reports Export)

- Priority is low.

# Project Management Approach

- Not changed but proposed plan: Transferring our repository to the GitHub organization
- We now use the PM's personal repository
- Members' user privileges are limited
- We plan to discuss at the next meeting

# Key Risks & Mitigation Strategies Update

- No additional risks identified
- Discord is strongly leveraged for consistent and timely communication during development
- Members continue to offer support and ideas proactively
- Use of GitHub organization to provide broader team access to repository settings

# Quality Assurance Metrics

## Defects — reported / fixed / open

38

### Reported

confirmed via failing tests, ~20 root causes

0

### Fixed

logged for team triage this sprint

38

### Open

100% — none remediated yet

## Test suite results — 386 scenario-tests



■ Passed **299**

■ Not yet built **49**

■ Known defect (xfail) **38**

151 pre-existing unit tests + 235 new acceptance tests, one run.

## Top open defects — regardless of demo timing

**CRITICAL** **Damage reports flag the wrong user** — increments the owner's damage counter instead of the borrower's who had the tool.

**CRITICAL** **Suspended members can't log in** — rejected outright instead of shown the required suspension notice.

**SERIOUS** **1-day reservations always rejected** — `start_date == end_date` should be a valid rental.

**SERIOUS** **Zero Hawaii Standard Time handling** anywhere in the backend, despite being an explicit ADR (ADR-006).

## Not yet implemented (separate from defects)

Messaging, listing reports & admin review, category management

Moderation CSV reports, admin reservations overview

Public member profiles

49 scenarios across 9 user stories — matches the Icebox/deferred stories from Slide 5 & 10



# Milestones

## Missed/Postponed

### 10 deferred User Stories (of the 35 created)

- Deferred user stories covered additional functionality for users and mostly admin tools
- Narrowed scope for initial launch and R1 demo

## Upcoming

### Testing and Version 2 Release

- **July 14:** Manual test cases created in addition to CI/CD workflow
- **July 27:** All deferred user stories included in updated release. Potential bug fixes after additional testing.

### Documentation Compilation

- **Aug 6:** Deployment Guide; updated GitHub
- **Showcase**
- **Aug 13:** Final presentation and demo

# Next Steps

1

## Deferred User Stories

### Functionality

- Timezone normalization
- Reservation messaging
- Notification center UI
- Member: Reporting a listing

### Admin Tools

- Review reports
- Track violations
- Suspend/reactivate a member
- Moderation history

2

## Improvements

### Usability

- Pagination on all list endpoints
- Advanced search filters

### R1 Retrospective

- Bug fixes for potential issues identified after R1

3

## TBD/Icebox

### Additional Admin Tools

- Managing tool categories (CRUD API)
- Generate community moderation reports

# Neighborhood Tool Sharing Demo

- Admin Invitation of New User, Tool Deactivation/Reactivation, Member Sign-up

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## Neighborhood Tool Sharing

Dashboard

Browse Tools ▾

Reservations

Review History

Demo Owner ▾

Available Tools

Returned Tools

Add New Tools

My Listings

R1 WORKFLOW

### Member Dashboard

Welcome, Demo Owner. Quick access to your tools, reservations, and notifications.

6

Tools in the system

0

My active reservations

0

My completed or closed

0

Unread notifications

#### My Tools

View available tools, browse listings, create new tool listings, and open tool detail pages.

[View tools](#)

#### My Reservations

Review borrower and owner reservations and take workflow actions like approve, deny, confirm pickup, and confirm return.

[View reservations](#)

#### Review History

See reviews you have given and received for returned reservations.

[View review history](#)

Thank you!

Questions?